A PROJECT TITLE

ON

Diabetes Risk Prediction Using Logistic Regression and Random Forest with Cross-Validation

project report submitted in partial fulfilment of the requirements for the Course

BACHELOR OF TECHNOLOGY

in

A.Y.: 2026-27

by

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1. Abstract:

Diabetes is a common chronic disease that requires early detection to reduce the risk of serious health complications. This project aims to predict the risk of diabetes using machine learning techniques based on patient health data from the Pima Indians Diabetes Dataset, which includes features such as glucose level, blood pressure, BMI, insulin level, age, and pregnancies. Two supervised machine learning algorithms, Logistic Regression and Random Forest, are used to classify whether a person is likely to have diabetes. To improve the reliability of the results, k-fold cross-validation is applied during model evaluation. The performance of both models is compared using evaluation metrics such as accuracy, precision, recall, and F1-score. This project demonstrates how machine learning can assist in the early prediction of diabetes, helping healthcare professionals make informed decisions and supporting timely medical intervention. Although it is not a substitute for clinical diagnosis, it serves as a useful decision-support tool for improving healthcare outcomes.

2. INTRODUCTION:

Diabetes is a chronic disease that affects millions of people worldwide and can lead to serious health complications if not detected early. This project uses machine learning to predict whether a person is at risk of diabetes based on medical data such as glucose level, blood pressure, BMI, age, and insulin level. Two classification algorithms, Logistic Regression and Random Forest, are implemented and evaluated using k-fold cross-validation to compare their performance and improve prediction reliability. The goal of this project is to develop an accurate diabetes risk prediction model that can assist healthcare professionals in early diagnosis and support better medical decision-making.

3. SOFTWARE REQUIREMENTS:

- Python – Programming language used to develop the machine learning model.
- Google Colab – Platform used to write, run, and test Python code.
- Visual Studio Code (VS Code) – Code editor for developing and managing the project.
- Pandas – Used for loading, cleaning, and preprocessing the dataset.
- CSV Dataset (Pima Indians Diabetes Dataset) – Used as the input data for training and testing the machine learning models.

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